

Python: Environment and Settings

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Course Info

- Thanks for all of your cooperation in forming the teams and providing available time slots 😊
 - Nice to see so many individual teams!
- Please check your team members at moodle
- TA time slots will be announced very soon
- **HW0** is out, the deadline is **9/23 (Sunday) 23:59**
 - HW0 is for individual! (每個人都必須獨自繳交)
- HW1 will be announced next week

What is Code? A Program?



食材

餅皮	...	蜂蜜	15g
低筋麵粉	100g	全蛋液（不含蛋殼）	28g
奶油	28g	鹽	1g
糖粉	30g	高筋麵粉（當手粉）	適量
奶粉	10g	蛋黃液（刷表面用）	適量

內餡

棗泥	1斤
綜合堅果（烤熟）	150g
葡萄乾（泡酒瀝乾）	80g



- 1** 把餅皮的材料混合，用拌壓的方式成團後，放入塑膠袋入冰箱冷藏30分即可使用



- 2** 葡萄乾泡酒瀝乾和烤熟的雜糧拌入棗泥/中備用



- 4** 把包好內餡的麵團放模型中，放在烤盤上輕輕壓一下，再往下推出



- 5** (1) 烤箱: 200度烤10分 (預熱)
(2) 烤箱: 190度烤10分
(3) 拿出刷蛋黃液2次
(4) 烤箱: 190度烤15分



- 3** 餅皮20g搓圓壓扁 內餡45g搓圓 餅皮包入內餡收口搓成陀圓形，要放入壓花那邊沾少許高筋麵粉

- 6** 月餅出爐後待涼放入保鮮盒，放室溫待2天後餅皮回油，會變得更柔軟好吃

What is Code? A Program?

Für Elise

L.V. Beethoven

Poco moto

pp

8

1. 2.

14

20

1. 2.

26

dolce

32

p



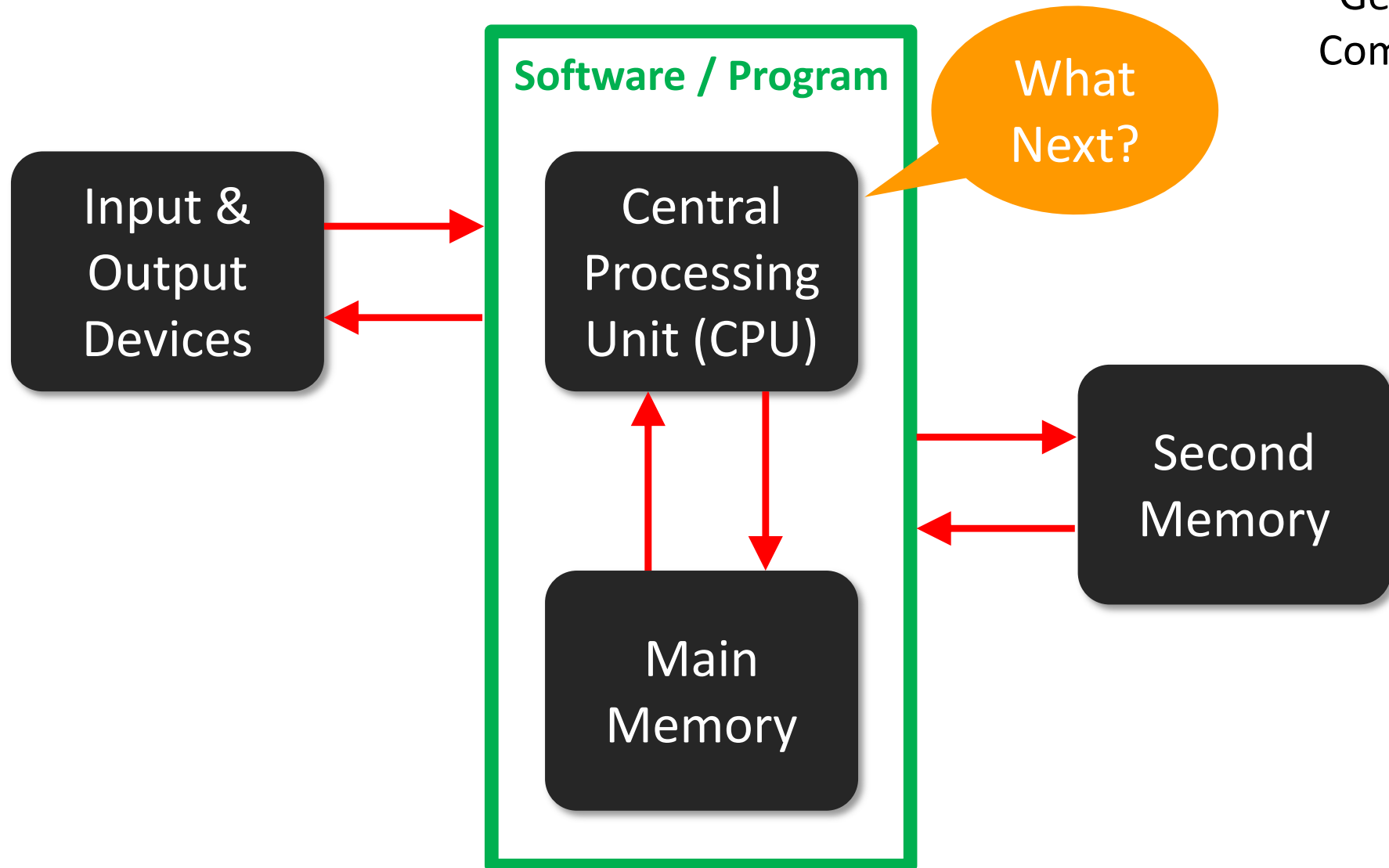
- Step 1 先踏併步往你的右邊
- Step 2 左手插腰
右手伸直握拳
從下方往上畫圈圈
- Step 3 同時左腳往左移動踏併步4拍
- Step 4 換手往上畫圈圈
踏併步4拍
回到原點
- Step 5 踩踢腳 4拍
換腳
再一次踩踢腳 4拍
- Step 6 再一次

What is Code? A Program?

- A **sequence** of **stored instructions**
 - It is a little piece of our intelligence in the computer
 - It is a little piece of our intelligence we can give to others - we figure something out and then we encode it and then give it to someone else to save them the time and energy of figuring it out
- A piece of **creative art**
 - Particularly when we do a good job on user experience

Basic Computing Architecture

General
Computer

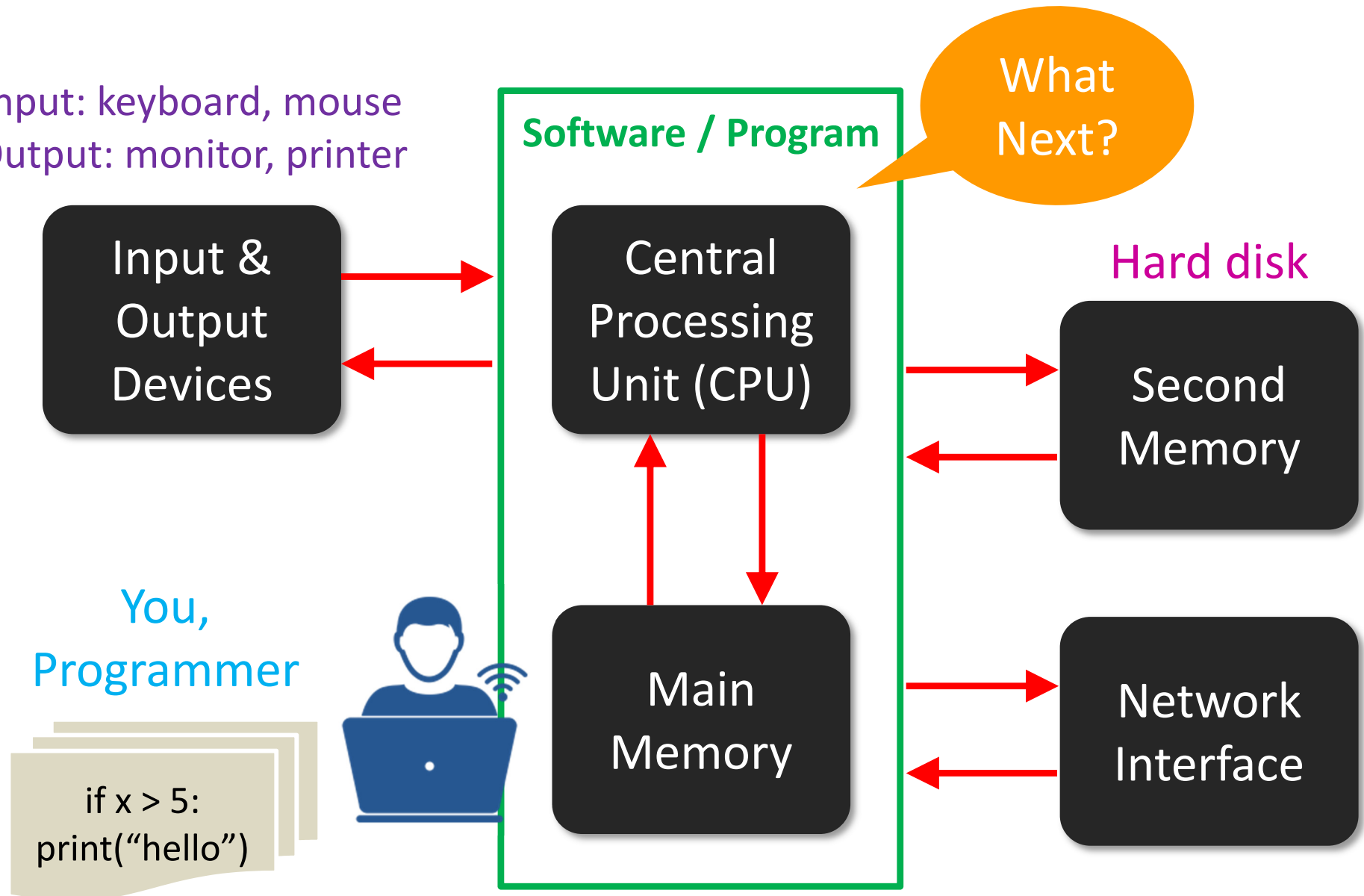


Basic Computing Architecture

- **Central Processing Unit (CPU):** Runs the Program
 - The CPU is always wondering “what to do next”?
 - Have the overall control of the computer and carries out instructions of computer program(s)
 - Not the brains exactly - very dumb but very very fast
- **Input and Output Devices**
 - Input: Keyboard, Mouse, Touch Screen
 - Output/Display: Screen, Speakers, Printer, DVD Burner
- **Main Memory**
 - Store programs and data files
 - Fast small temporary storage
 - Lost on reboot (Random Access Memory, RAM)
- **Secondary Memory**
 - Store programs and data files
 - Slower large permanent storage
 - Lasts until deleted – disk drive / memory stick
- **Network Interface**
 - Allow the exchange of data with other computers

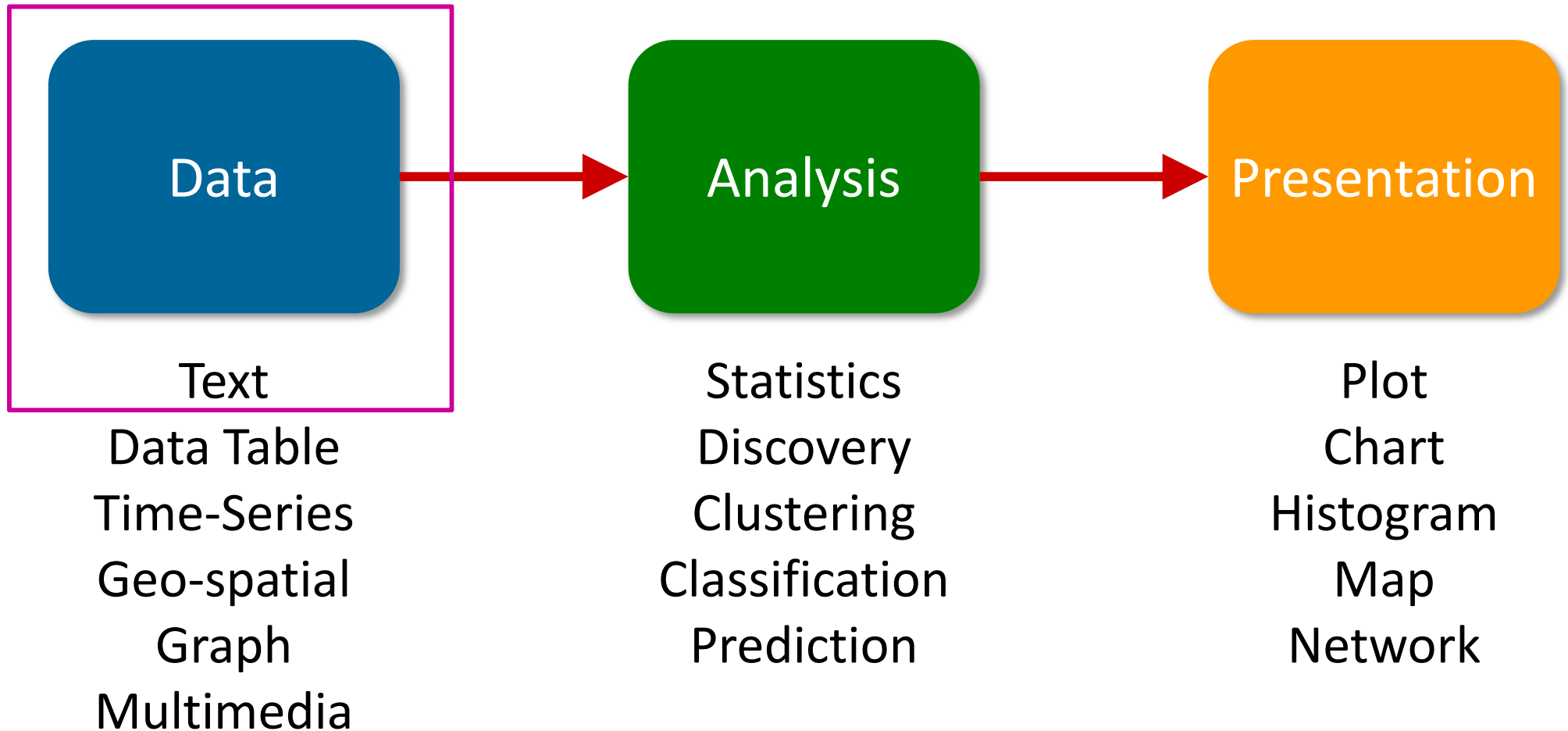
Basic Computing Architecture

Input: keyboard, mouse
Output: monitor, printer



Python Programming for Data Analysis

This/Next Semester



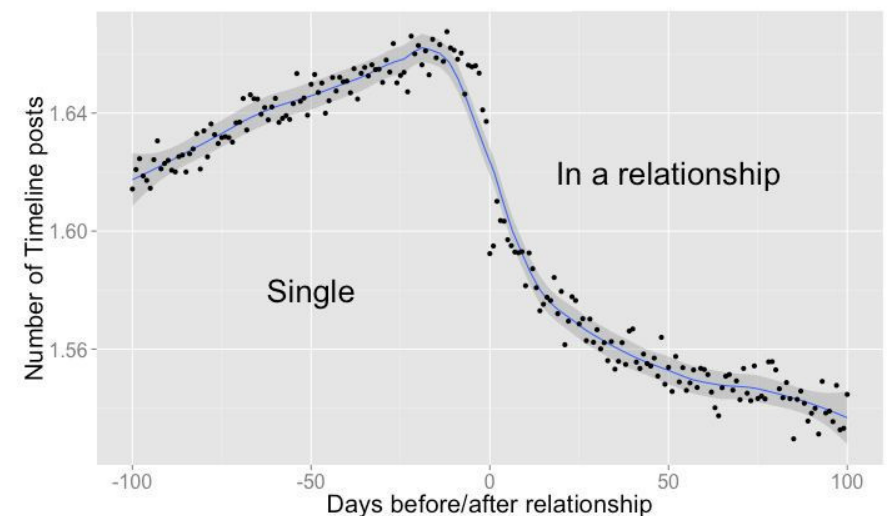
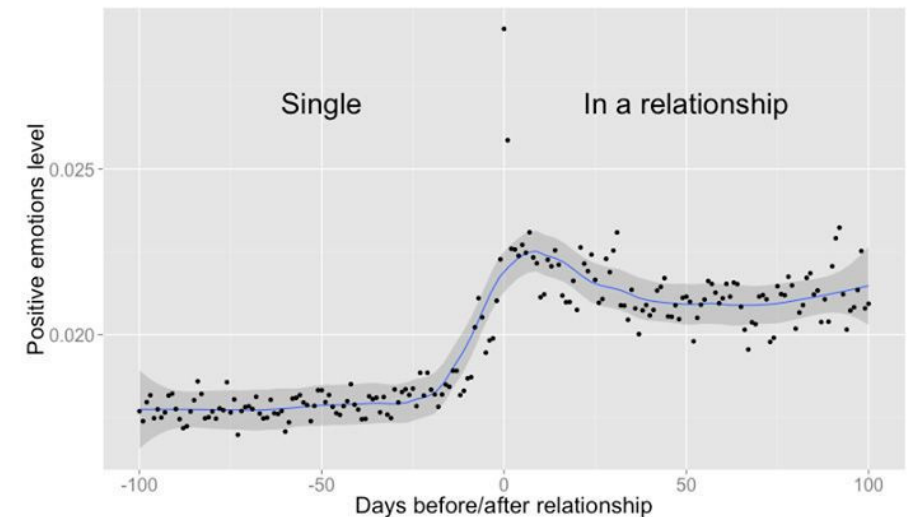
Python Programming for Data Analysis



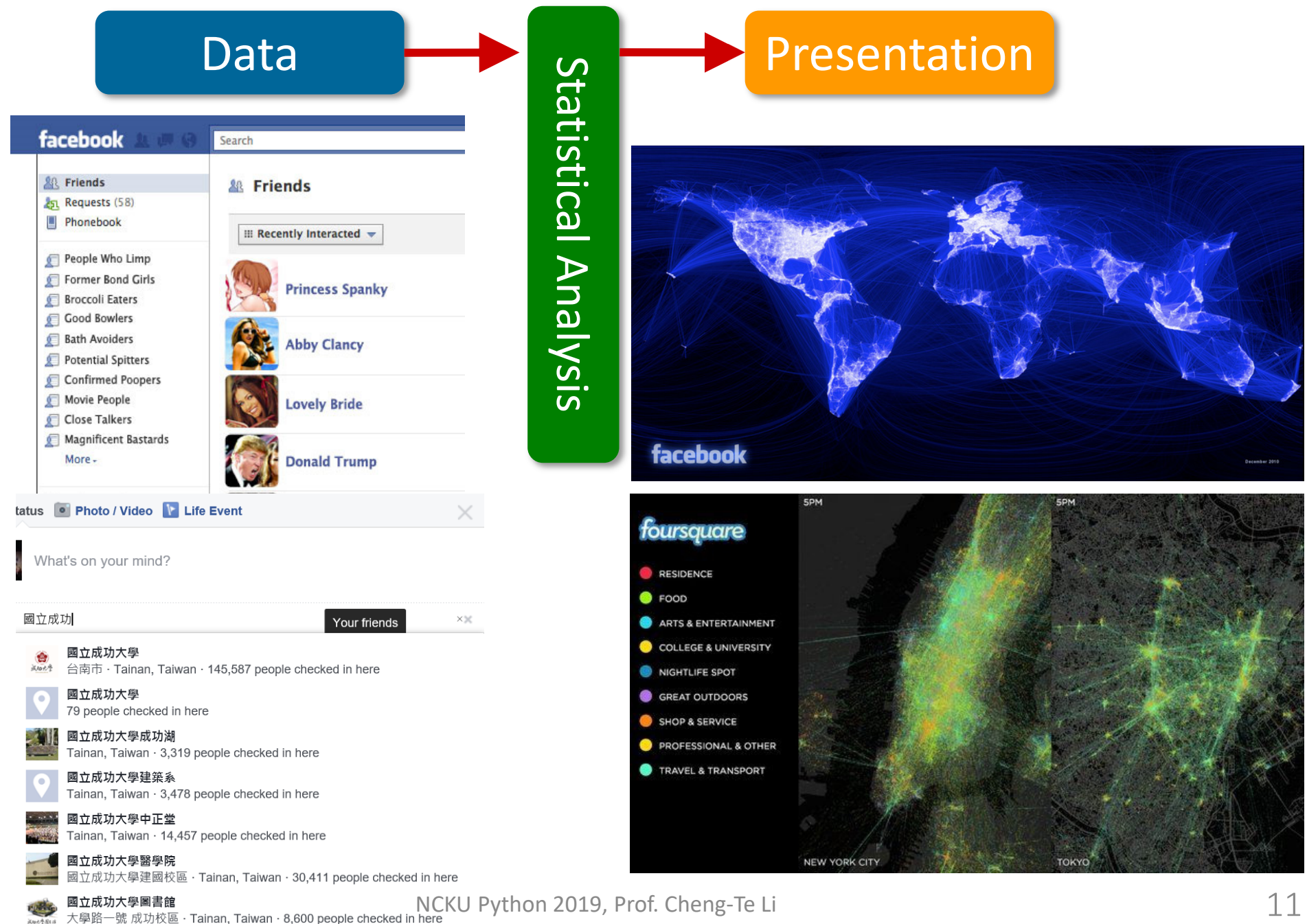
Data

Statistical Analysis

Presentation



Python Programming for Data Analysis



Python: The First Step

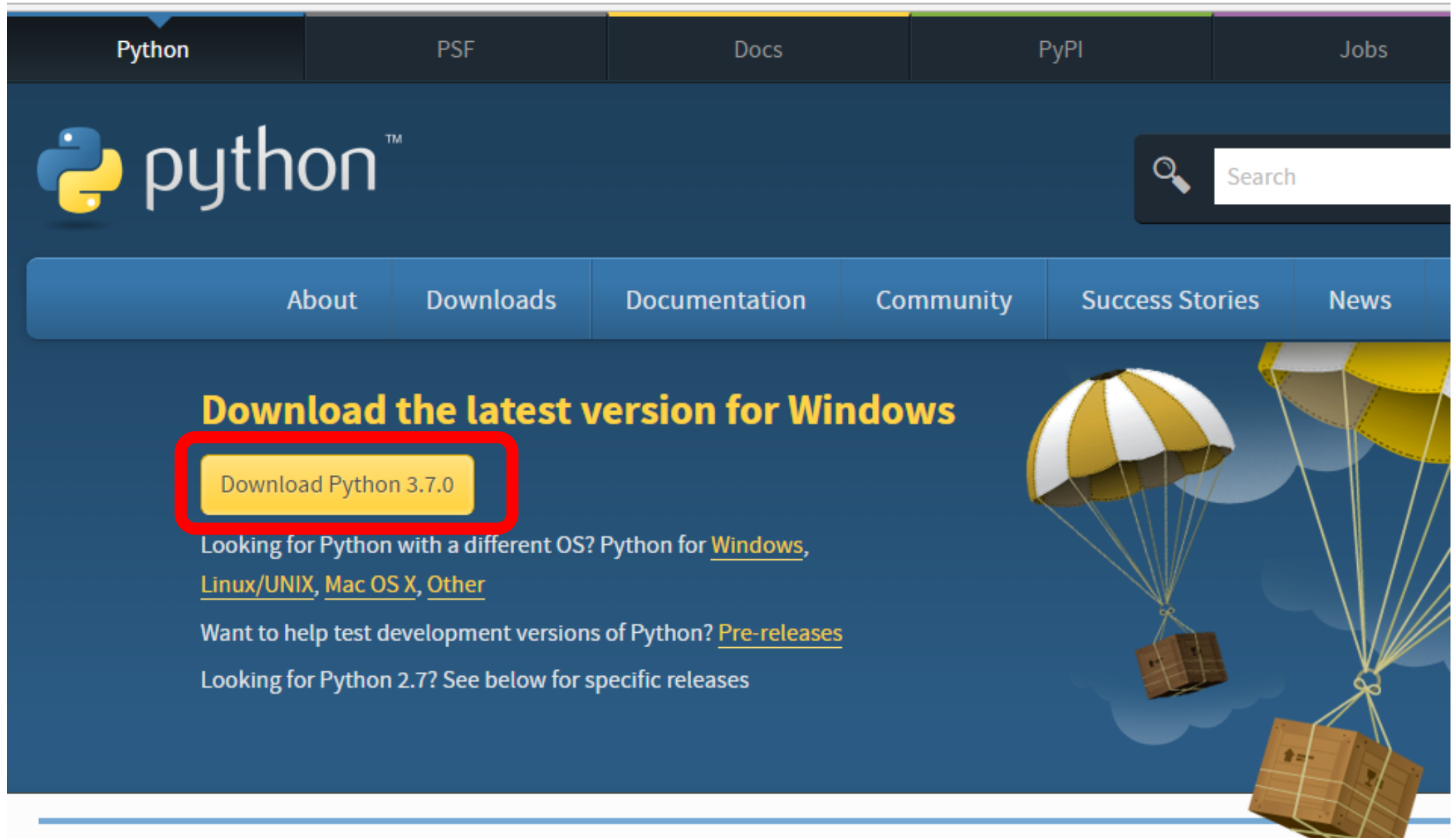


Download Python: <https://www.python.org/> (**Windows**, OS X, Linux)
Text Editor: Python IDLE, 記事本, **Sublime Text**, Notepad++, Vim, Pydev, Spyder, IPython Notebook, PyCharm, etc.)

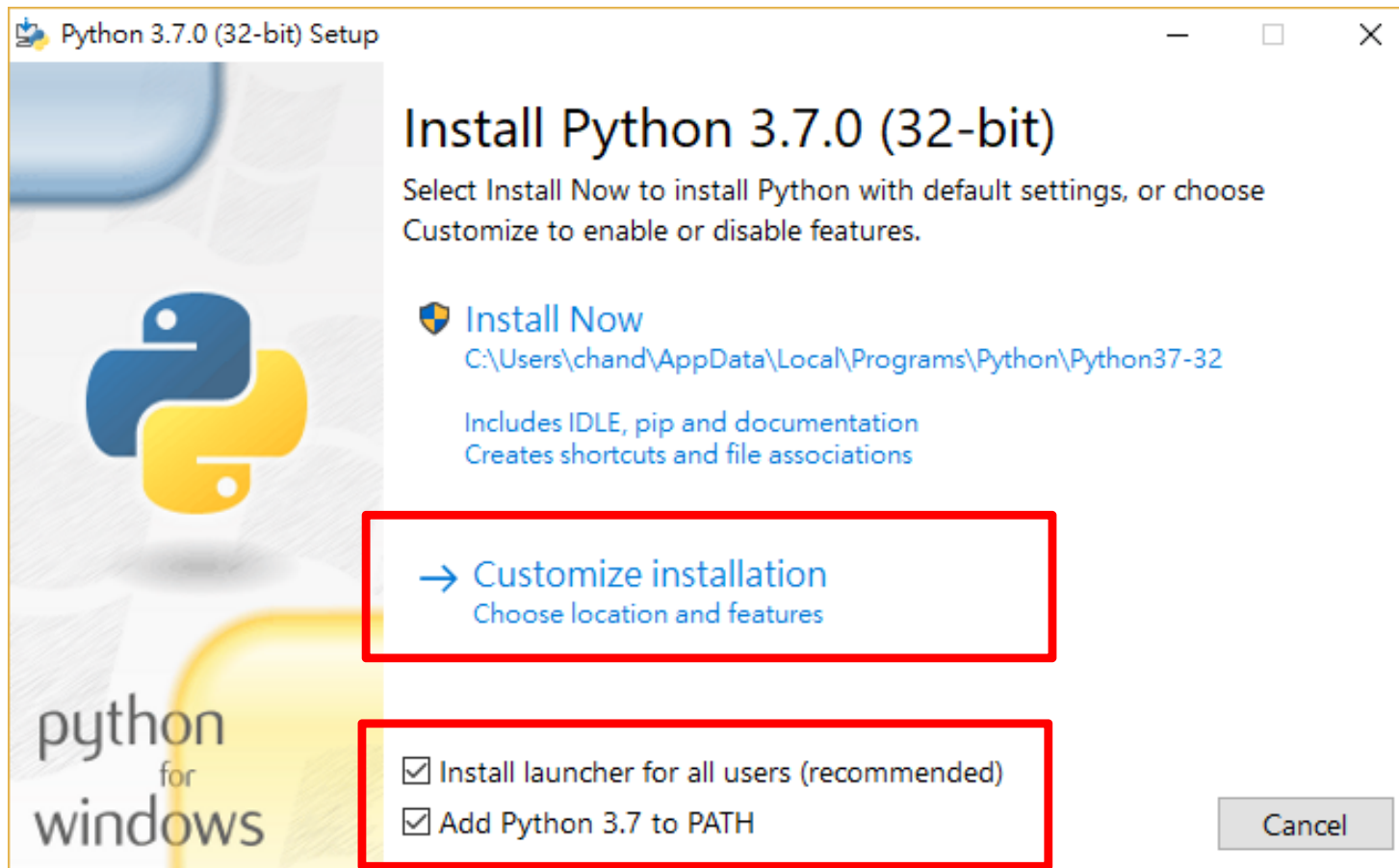
Downloads: Python

<https://www.python.org/downloads/>

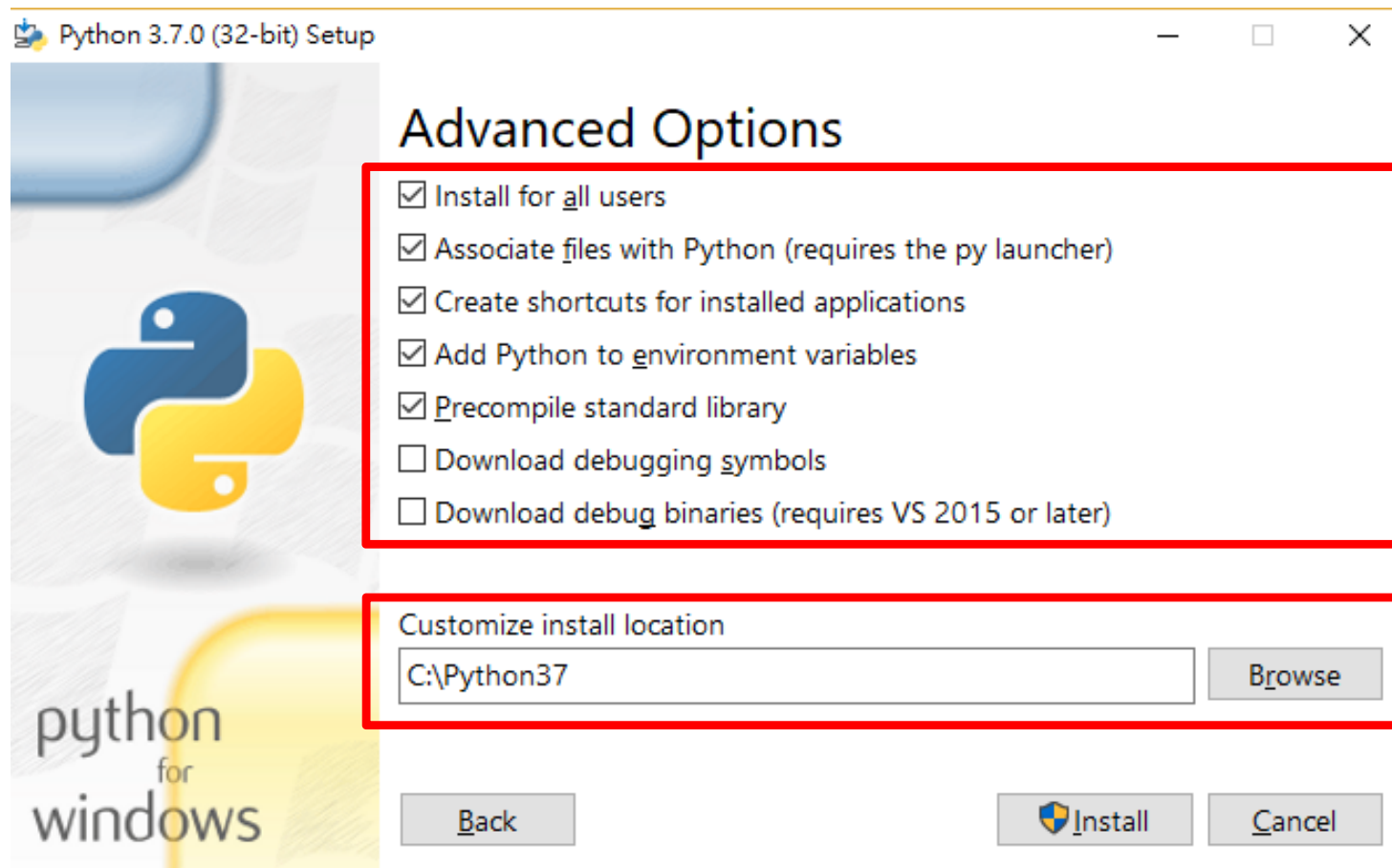
<https://www.python.org/downloads/>

The image is a screenshot of the Python.org website's 'Downloads' section. At the top, there is a navigation bar with links for 'Python', 'PSF', 'Docs', 'PyPI', and 'Jobs'. Below this is a dark blue header featuring the Python logo and a search bar. A secondary navigation bar contains links for 'About', 'Downloads', 'Documentation', 'Community', 'Success Stories', and 'News'. The main content area has a dark blue background with a large yellow and white striped parachute carrying a crate, and another yellow parachute with a crate below it. The text 'Download the latest version for Windows' is prominently displayed in yellow. Below this, a yellow button with the text 'Download Python 3.7.0' is highlighted with a red rectangular border. Further down, there are links for 'Python for Windows, Linux/UNIX, Mac OS X, Other', 'Pre-releases', and a note about Python 2.7.

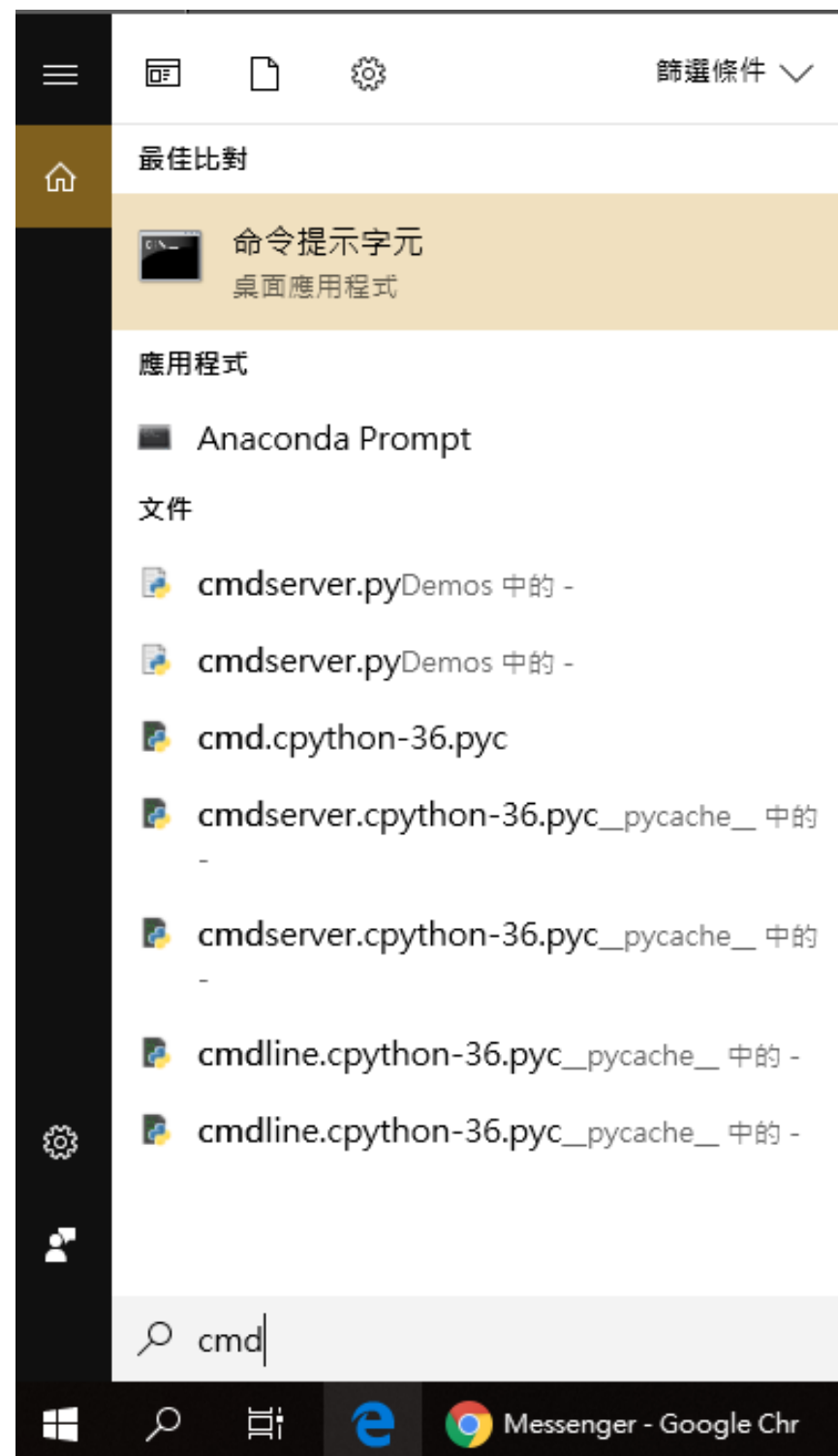
Install Python 3.7.0



Install Python 3.7.0



- 點選開始鍵，在“搜尋”欄(放大鏡圖示)中打上cmd，會出現**命令提示字元**，按下Enter (或按快捷鍵:windows 鍵+R，輸入cmd)



- cmd會呼叫出視窗(此為"命令提示字元"視窗)，在視窗內輸入"**python -V**" (注意"V"為大寫)
- 若下方輸出"Python 3.7.0" 則代表python安裝成功，即可關閉視窗



```
CA. 命令提示字元
Microsoft Windows [版本 10.0.17134.228]
(c) 2018 Microsoft Corporation. 著作權所有，並保留一切權利。
C:\Users\chand>python -V
Python 3.7.0
C:\Users\chand>
```

Successful Installation

```
命令提示字元 - python
Microsoft Windows [版本 6.3.9600]
(c) 2013 Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\Chandler>python
Python 3.5.2 (v3.5.2:4def2a2901a5, Jun 25 2016, 22:01:18) [MSC v.1900 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

主提示符號
(primary prompt)

What Next?
接下來要做什麼?

Python Interpreter (Python直譯器)

計算機

關閉直譯器: Ctrl + c 或 quit()

Sublime Text 3 <https://www.sublimetext.com/3>

Sublime Text

Download

Buy

News

Forum

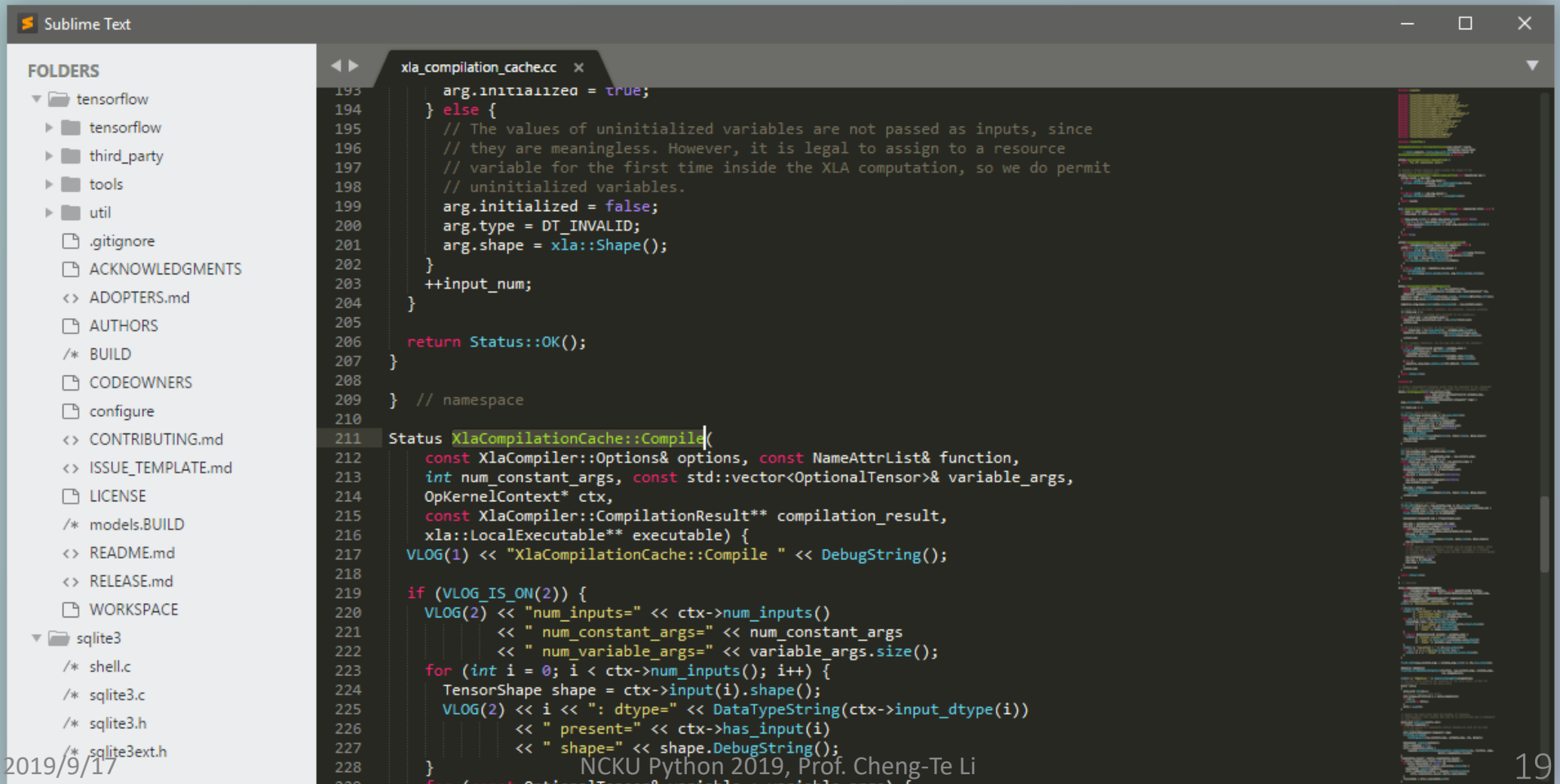
Support

A sophisticated text editor for code, markup and prose

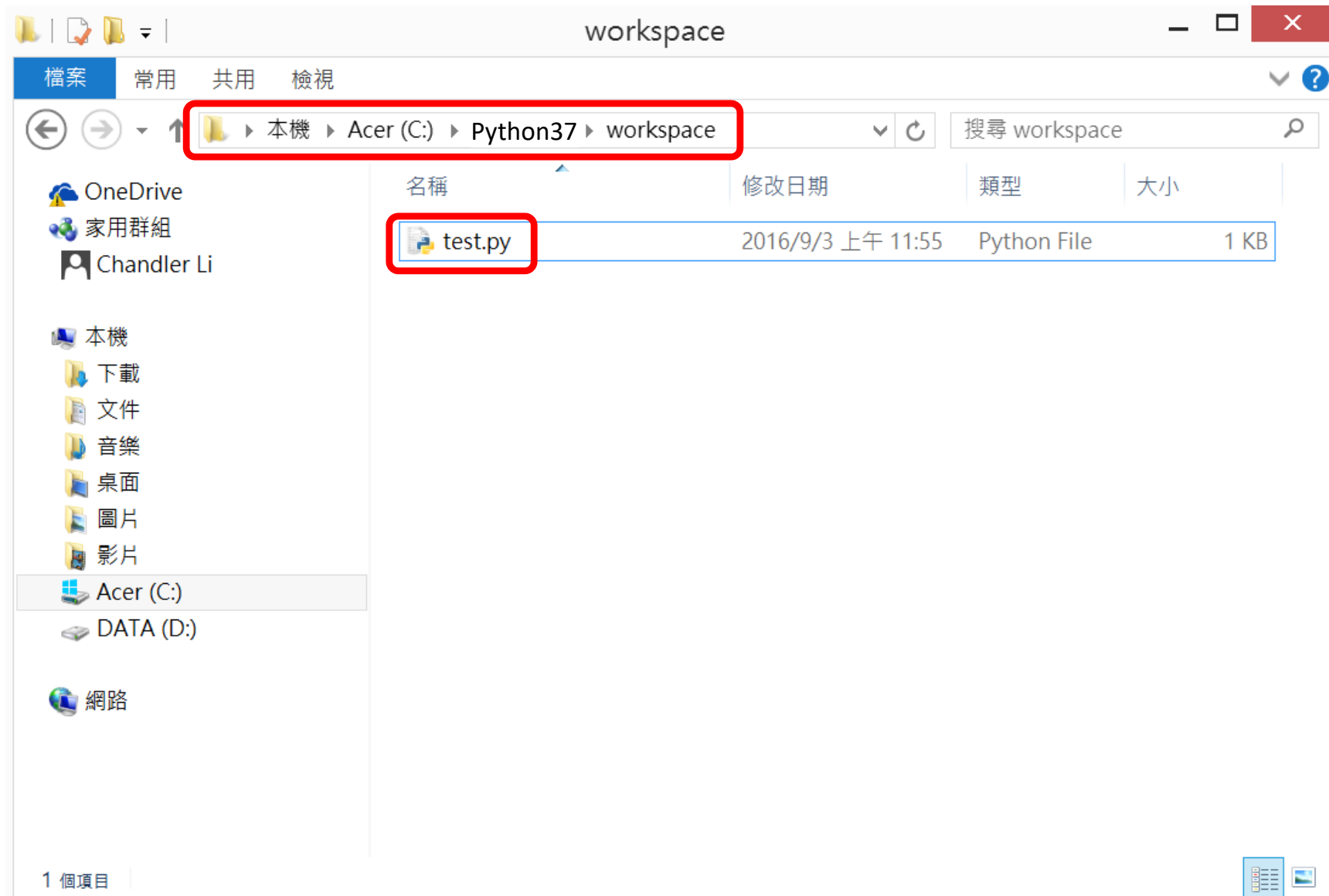
DOWNLOAD FOR WINDOWS

Sublime Text 3 (Build 3143)

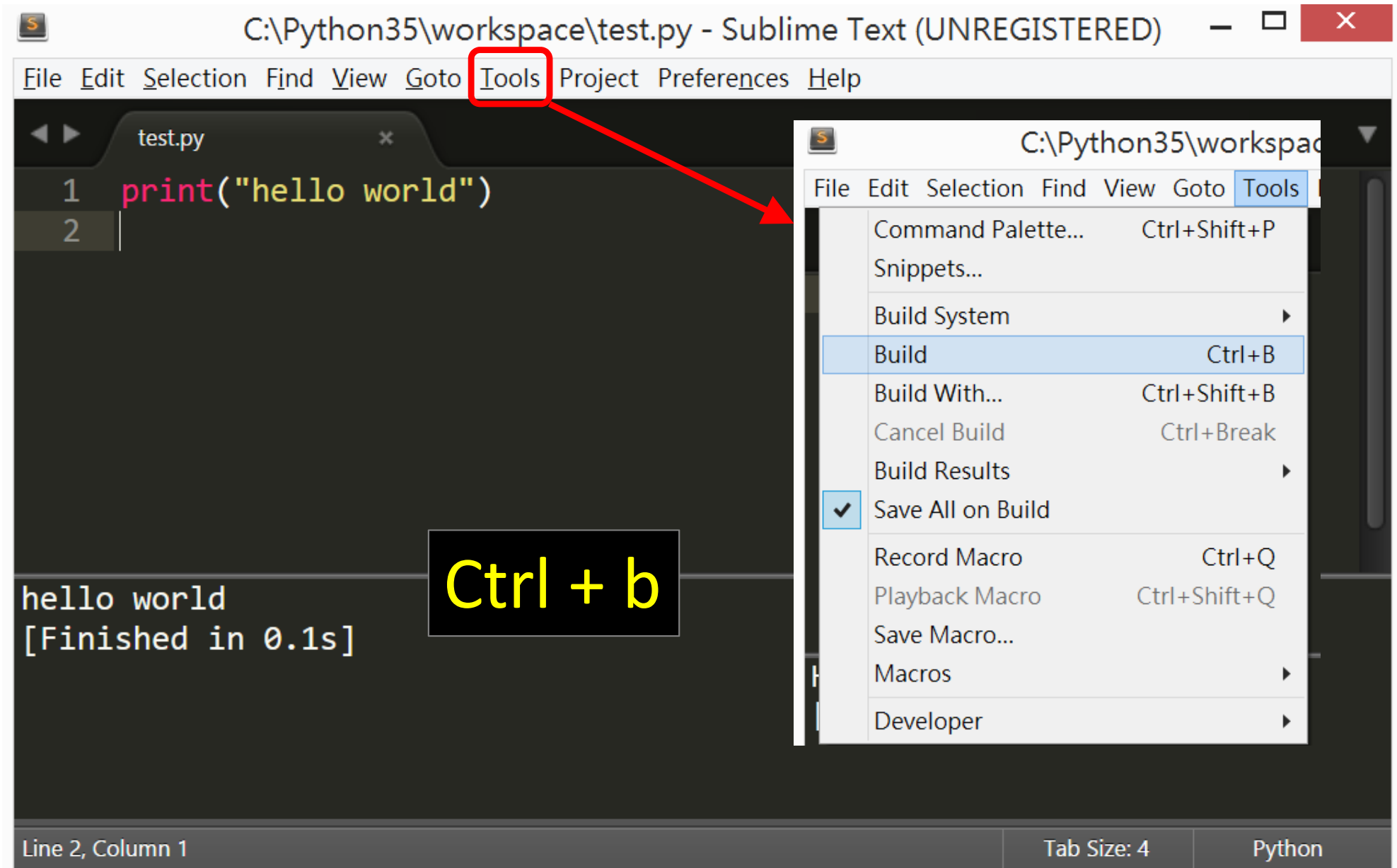
[Changelog](#)



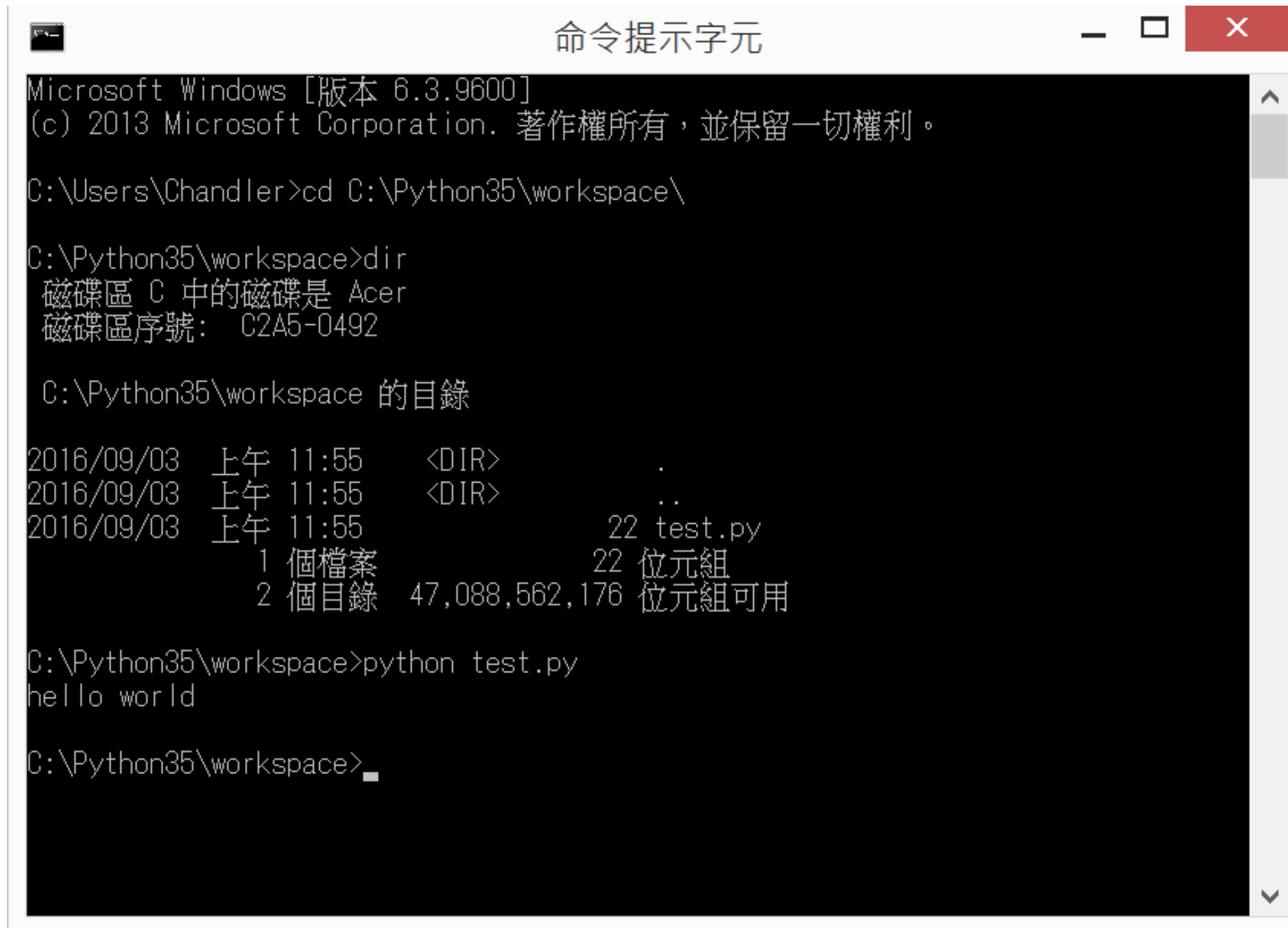
Create “workspace” and .py file



Sublime Text Editor



Execute Python Program



```
Microsoft Windows [版本 6.3.9600]
(c) 2013 Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\Chandler>cd C:\Python35\workspace\

C:\Python35\workspace>dir
磁碟區 C 中的磁碟是 Acer
磁碟區序號: C2A5-0492

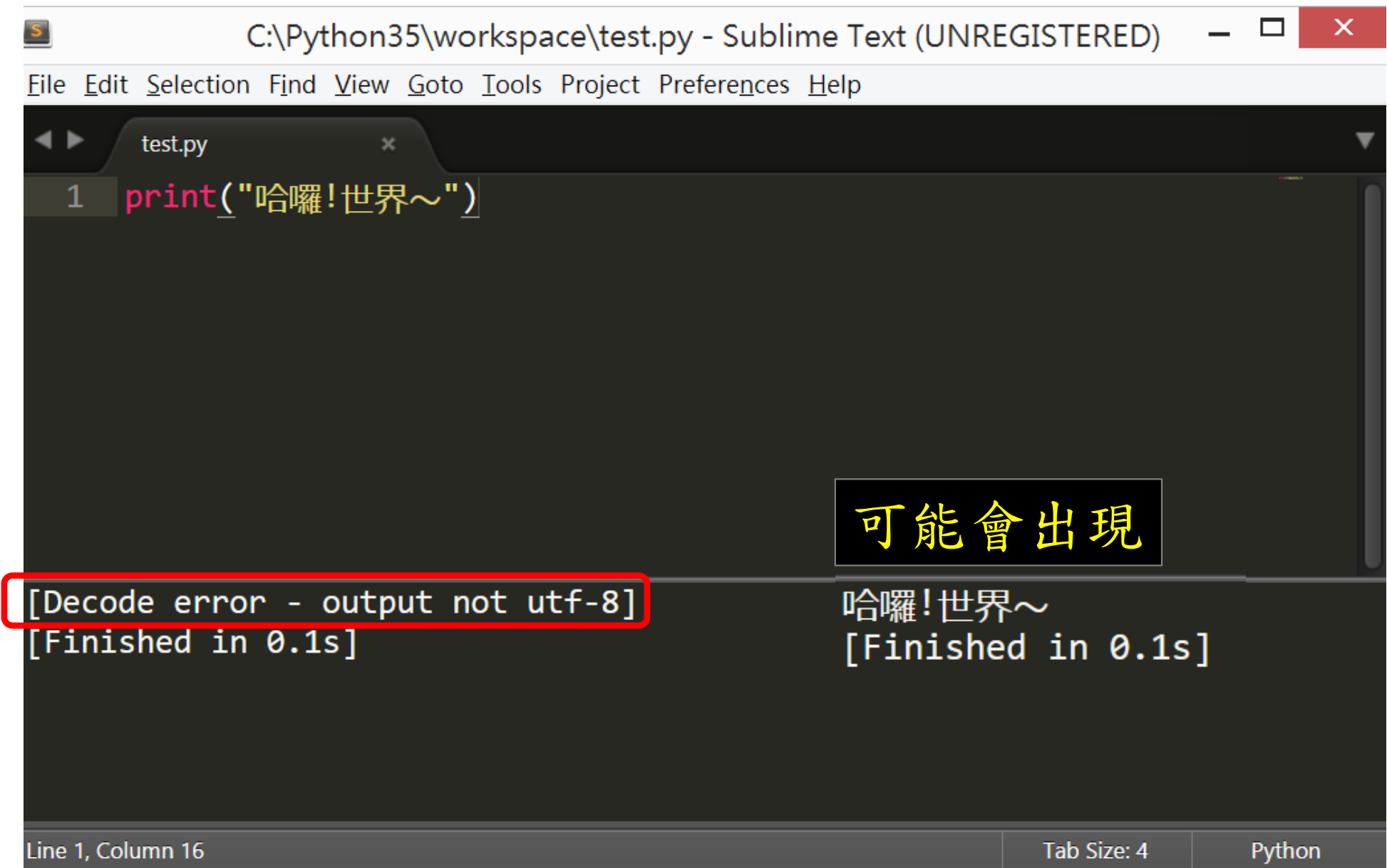
C:\Python35\workspace 的目錄

2016/09/03 上午 11:55    <DIR>          .
2016/09/03 上午 11:55    <DIR>          ..
2016/09/03 上午 11:55                22 test.py
                1 個檔案                22 位元組
                2 個目錄    47,088,562,176 位元組可用

C:\Python35\workspace>python test.py
hello world

C:\Python35\workspace>
```

How about Chinese?



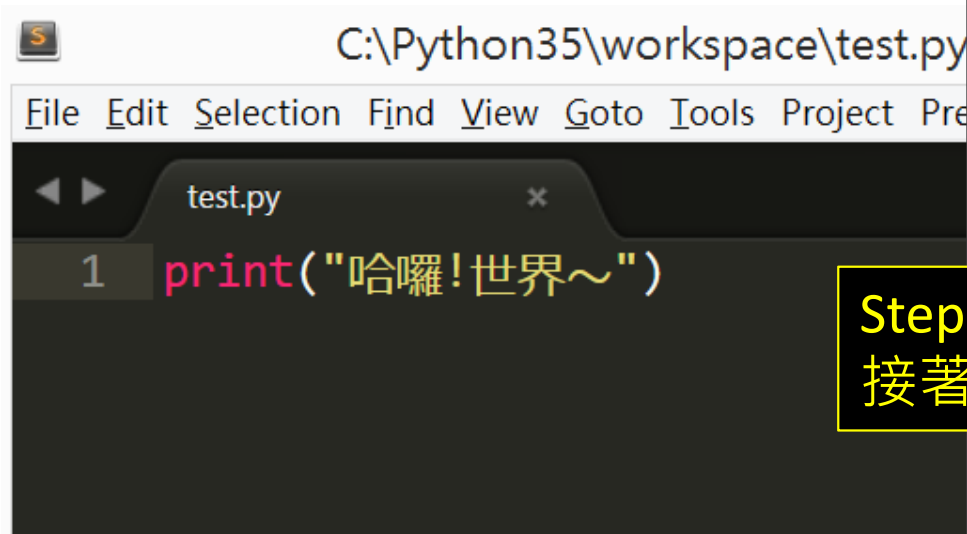
The screenshot shows the Sublime Text editor window titled "C:\Python35\workspace\test.py - Sublime Text (UNREGISTERED)". The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The editor has a single tab for "test.py" containing the following code:

```
1 print("哈囉!世界~")
```

Below the editor, the console output is displayed. The first line, "[Decode error - output not utf-8]", is highlighted with a red rectangle. The second line is "[Finished in 0.1s]". To the right of the console output, the text "可能會出現" (Might appear) is shown in a yellow box, followed by the output "哈囉!世界~" and "[Finished in 0.1s]". The status bar at the bottom indicates "Line 1, Column 16", "Tab Size: 4", and "Python".

Package Control Installation for Sublime

<https://packagecontrol.io/installation>



**Step 1: 打開連結這個網頁，
接著找到Sublime Text 3這段程式碼。**

**Step 2: 從選單的View > Show Console。
或是按快捷鍵「ctrl+`」，開啟命令列。**

```
Running python -u "C:\Python35\works  
Writing file /C/Python35/workspace/t  
Running python -u "C:\Python35\works  
Running py
```

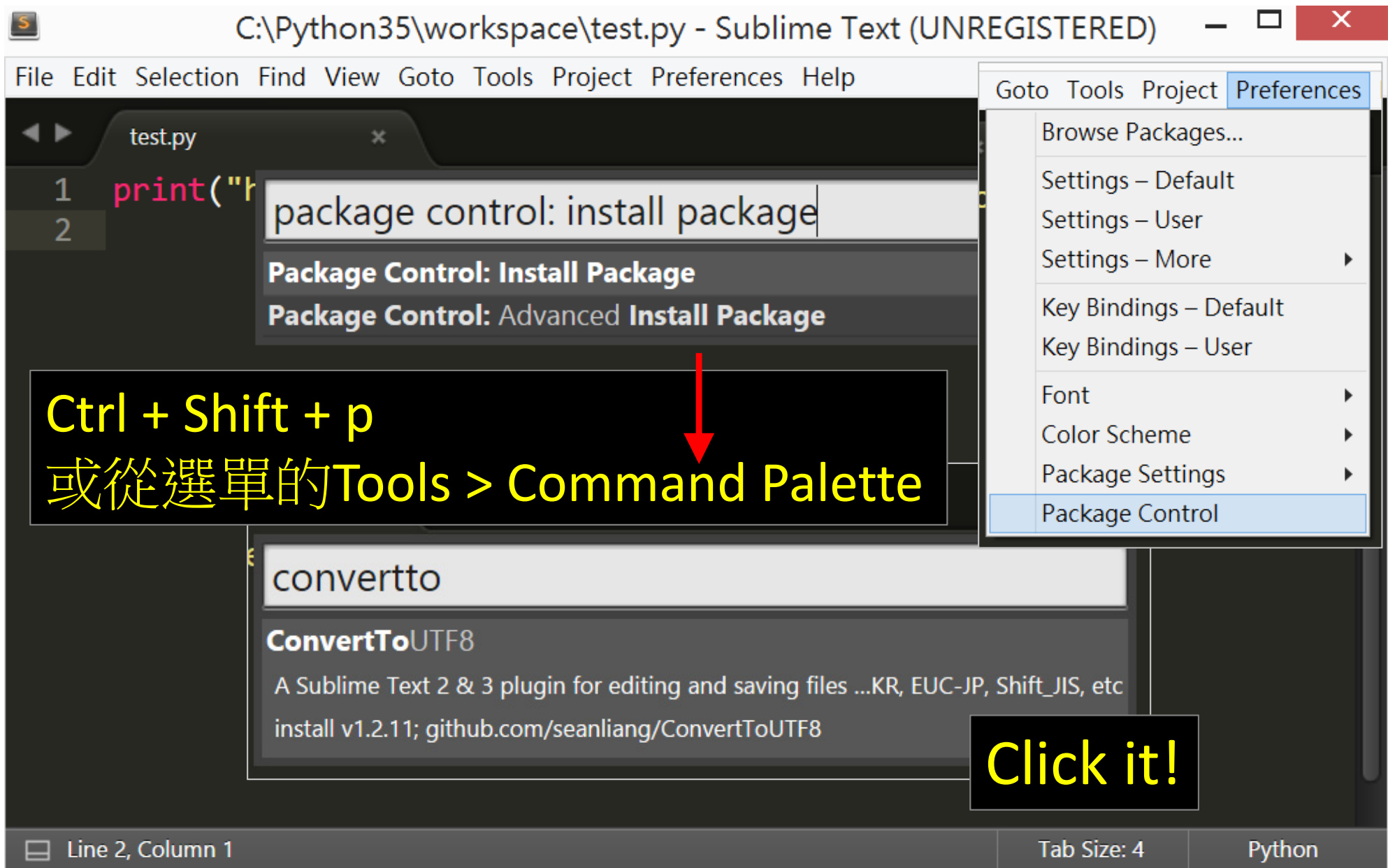
Step 3: 複製貼上

```
h)) if dh != h else open(os.path.join( ipp, pf), 'w
```

Step 4: 按Enter



Install Sublime Packages



In-class Exercise

- Do by yourself!
 - Install Python
 - Install Sublime Text 3
 - Install Package Control
 - Install Convert2UTF8